

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Technology (CSE)

QUIZ #2 (SET-SAMPLE)
DURATION: 25 MINUTES

SUMMER SEMESTER, 2024-2025
FULL MARKS: 20

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID: _____

1. For a binary logistic regression model,

$$h_{\theta}(x) = \sigma(\theta^T x),$$

which condition is equivalent to predicting class 1 using a threshold of 0.5?

- A. $\theta^T x < 0$
- B. $\theta^T x \geq 0$
- C. $\theta^T x \geq 0.5$
- D. $\theta^T x = 1$

2
(CO1)
(PO1)

Solution:

Since $\sigma(z) \geq 0.5$ if and only if $z \geq 0$, the model predicts class 1 exactly when $\theta^T x \geq 0$.

2. Why is binary cross-entropy preferred over mean squared error for standard logistic regression?

- A. It removes the need for a sigmoid function.
- B. It guarantees zero training error.
- C. **It is the negative log-likelihood of the Bernoulli model and gives a convenient gradient.**
- D. It makes every decision boundary nonlinear.

2
(CO1)
(PO1)

Solution:

Binary cross-entropy follows from maximum likelihood under a Bernoulli output model. Together with the sigmoid, it simplifies the gradient to an error term of the form $(\hat{y} - y)x$.

3. Which expression is the vectorized gradient of the unregularized logistic regression loss for m examples, where $X \in \mathbb{R}^{m \times d}$?

- A. $\frac{1}{m}X(\hat{y} - y)$
- B. $\frac{1}{m}X^T(\hat{y} - y)$
- C. $\frac{1}{m}(\hat{y} - y)^T X^T$
- D. $X^T X \theta$

2
(CO1)
(PO1)

Solution:

The prediction error vector has dimension m , so multiplying it by $X^T \in \mathbb{R}^{d \times m}$ produces a gradient vector in $\mathbb{R}^d$:

$$\nabla_{\theta} J(\theta) = \frac{1}{m} X^T (\hat{y} - y).$$

4. What happens if all hidden layers of a neural network use only affine transformations and no nonlinear activation functions? 2
(CO2)
(PO1)
- A. The network can represent any continuous function.
 - B. The entire network is equivalent to a single affine transformation.**
 - C. Backpropagation becomes mathematically impossible.
 - D. The network automatically behaves like a nearest-neighbour classifier.

Solution:

A composition of affine transformations is still affine. Therefore, additional layers do not increase the expressive power unless nonlinear activation functions are inserted between them.

5. Which statement best explains why sigmoid activations can cause vanishing gradients in a deep network? 2
(CO2)
(PO1)
- A. The sigmoid derivative is always greater than 1.
 - B. The sigmoid output is unbounded.
 - C. Its derivative is at most 0.25, so repeated multiplication can rapidly shrink gradients.**
 - D. It is not differentiable at the origin.

Solution:

Since

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)) \leq 0.25,$$

multiplying many such derivatives during backpropagation may make the gradient extremely small in earlier layers.

6. Which statement correctly describes the Universal Approximation Theorem? 2
(CO2)
(PO1)
- A. A single neuron can exactly represent every continuous function.
 - B. Gradient descent always finds the optimal approximating network.
 - C. A one-hidden-layer network with enough neurons and a suitable nonlinear activation can approximate any continuous function on a compact domain arbitrarily well.**
 - D. Deep networks always require more parameters than shallow networks.

Solution:

The theorem is an existence result about representational capacity. It does not state how to find the parameters, how many neurons are needed, or whether the learned model will generalize.

7. A logistic regression model predicts $\hat{y} = 0.2$ for an example whose true label is $y = 1$. Compute the binary cross-entropy loss for this example. 2
(CO1)
(PO1)

Solution:

For $y = 1$,

$$L = -\log(\hat{y}) = -\log(0.2) \approx 1.609.$$

Rubric:

- 1 mark for selecting the correct loss expression.
- 1 mark for the correct numerical value.

8. Briefly compare L1 and L2 regularization in terms of their effects on model weights. 2
(CO1)
(PO1)

Solution:

L1 regularization adds a penalty proportional to $\sum_j |\theta_j|$ and can drive some weights exactly to zero, producing sparse models. L2 regularization adds a penalty proportional to $\sum_j \theta_j^2$ and generally shrinks all weights smoothly toward zero without making many of them exactly zero.

Rubric:

- 1 mark for correctly describing L1 regularization.
- 1 mark for correctly describing L2 regularization.

9. Explain why initializing all weights in a hidden layer to zero prevents the neurons in that layer from learning different features. 2
(CO2)
(PO1)

Solution:

With identical zero initialization, all neurons in the layer produce identical outputs and receive identical gradients. Their weights therefore remain equal after every update. This symmetry prevents the neurons from learning distinct features.

Rubric:

- 1 mark for identifying identical outputs or gradients.
- 1 mark for explaining the resulting symmetry problem.

10. For a neural-network output layer using a sigmoid activation and binary cross-entropy loss, state the simplified expression for $dZ^{[L]}$ and explain why this simplification is useful. 2
(CO2)
(PO1)

Solution:

The output-layer error is

$$dZ^{[L]} = A^{[L]} - Y.$$

The derivative terms from the sigmoid and the binary cross-entropy loss cancel, leaving a simple prediction-error expression. This makes backpropagation computationally efficient and easier to implement.

Rubric:

- 1 mark for the correct expression.
- 1 mark for a valid explanation of the simplification.